SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1703**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I chakradhar Peela 192412400 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Chakradhar peela (192412400)

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

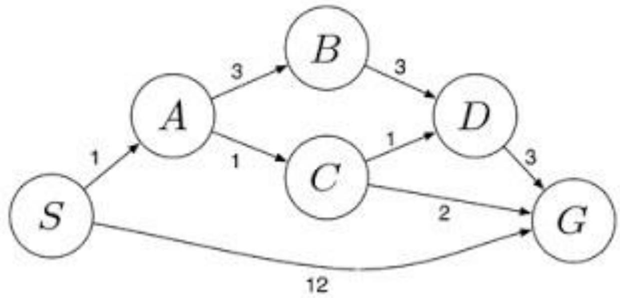
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Problem Understandin g	20%	Clearly interprets problem, identifies all parameters and	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
		requirements accurately			

Answers

1.Water Jug Problem

Problem:

- Jug A = 4 gallons
- Jug B = 3 gallons
- Goal: Get exactly 2 gallons in the 4-gallon jug.

State Representation

Represent the state as:

(4-Gallon Jug, 3-Gallon Jug)

Initially:

(0, 0)

Step-by-Step Calculation

Step	Action	4-Gallon Jug	3-Gallon Jug
Initial	Both jugs are empty	**0**	**0**
1	Fill the 3-gallon jug	**0**	**3**
2	Pour 3-gallon jug into 4-gallon jug	**3**	**0**
3	Fill the 3-gallon jug again	**3**	**3**
4	Pour water into the 4-gallon jug until it is full (needs only 1 gallon)	**4**	**2**
5	Empty the 4-gallon jug onto the ground	**0**	**2**
6	Pour the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug	**2**	**0**

Final State

(2, 0)

Hence, the **4-gallon jug contains exactly 2 gallons**, which is the required solution.

State Space Representation

Start
(0,0)

↓
Fill 3-gallon jug
(0,3)

↓
Pour into 4-gallon jug
(3,0)

↓
Fill 3-gallon jug
(3,3)

↓
Pour into 4-gallon jug
(4,2)

↓
Empty 4-gallon jug
(0,2)

↓
Pour remaining water into 4-gallon jug

(2,0) Goal Achieved
Answer for Assignment
Initial State: (0,0)
Goal State: (2,0)

2. **2. Mars Rover Agent – Analysis**
A **Mars Rover** is an intelligent autonomous agent that explores the surface of Mars, collects scientific data, analyzes rocks and soil, and sends information back to Earth.

i. What are the precepts for this agent?
Precepts are the inputs the rover receives from its sensors.
- Images from navigation and scientific cameras
- Rock and soil composition data
- Temperature readings
- Atmospheric pressure measurements
- Humidity and weather conditions

- Radiation levels
- Terrain information (rocks, slopes, sand dunes)
- Position and orientation (GPS-like localization using onboard navigation)
- Wheel movement and obstacle detection
- Battery and power status

ii. Characterize the operating environment

The Mars Rover operates in the following environment:

Environment Property	Description
Partially Observable	The rover cannot observe the entire Martian environment at once.
Deterministic/Stochastic	Mostly stochastic because dust storms, wheel slippage, and terrain uncertainties may occur.
Sequential	Each action affects future actions and mission progress.
Dynamic	Weather, lighting, and terrain conditions may change over time.
Continuous	Position, movement, and sensor values change continuously.
Single Agent	The rover performs its tasks independently.
Unknown	The rover explores areas that are not fully known beforehand.

iii. What are the actions the agent can take?

The Mars Rover can perform the following actions:

- Move forward, backward, left, or right
- Avoid obstacles
- Capture images and videos
- Collect rock and soil samples
- Drill into rocks
- Analyze samples using onboard scientific instruments
- Measure atmospheric conditions
- Store collected samples
- Transmit data and images to Earth
- Recharge batteries using solar panels (if available)

- Plan and adjust its exploration path

iv. How can one evaluate the performance of the agent?

The rover's performance can be evaluated using the following criteria:

- Successfully reaches assigned destinations
- Number of useful scientific samples collected
- Accuracy of sample analysis
- Quality of images and scientific data
- Safe navigation without collisions
- Efficient use of battery power
- Reliable communication with Earth
- Mission completion within the planned time
- Ability to operate autonomously in harsh conditions

v. What sort of agent architecture is most suitable for this agent? Why?

Most Suitable Architecture: Utility-Based Agent with Model-Based Features

Reason:

A Mars Rover must make intelligent decisions in an uncertain and changing environment. A Utility-Based Agent is the most suitable because it:

- Evaluates multiple possible actions before choosing the best one.
- Maximizes mission success while minimizing risks.
- Uses an internal model of the environment to predict outcomes.
- Adapts to unexpected obstacles, rough terrain, and changing weather.
- Balances objectives such as safety, energy conservation, and scientific exploration.

Since Mars has an unpredictable environment and communication delays with Earth, the rover must make many decisions autonomously. Therefore, a Utility-Based Agent combined with a Model-Based architecture provides the best balance between autonomous decision-making, safety, and efficient scientific exploration.

3. **Formulate the 8-Queens Problem Using Search Strategies**

Problem Statement

The 8-Queens Problem is to place 8 queens on an 8×8 chessboard such that no two queens attack each other.

A queen can attack another queen if they are in the:

- Same row
- Same column

- Same diagonal

The objective is to find a valid arrangement where all 8 queens are placed safely.

Problem Formulation

1. Initial State

- An empty 8 × 8 chessboard.
- No queens are placed.

State:

Empty Board

2. State Space

- Each state represents a partial arrangement of queens.
- One queen is placed in each column.
- The search continues until all 8 queens are placed safely.

Example:

```
Q . . . . . . .
. . . . . . . .
. Q . . . . . .
. . . . . . . .
. . . Q . . . .
. . . . . . . .
. . . . . Q . .
. . . . . . . .
```

3. Actions (Operators)

The agent can perform the following action:

- Place one queen in a safe row of the next column.
- If no safe position exists, backtrack and try another position.

Possible actions:

- Place Queen in Row 1
- Place Queen in Row 2
- ...
- Place Queen in Row 8

4. Goal Test

The goal is achieved when:

- Exactly **8 queens** are placed.
- No two queens attack each other.

That means:

- No queens share the same row.
- No queens share the same column.
- No queens share the same diagonal.

5. Path Cost
Each queen placement costs **1**.
Total path cost for a complete solution:
Cost = 8
since eight queens are placed.

Search Strategy
The most common search strategy used is Backtracking Search (Depth-First Search).
Steps

1. Start with an empty board.
2. Place the first queen in Column 1.
3. Move to Column 2 and place another queen safely.
4. Continue placing queens column by column.
5. If no safe position is available:
   - Remove the last queen (Backtrack).
   - Try another row.
6. Repeat until all 8 queens are safely placed.

Example Solution
One valid arrangement is:

Column	Row
1	1
2	5
3	8
4	6
5	3
6	7
7	2
8	4

Board Representation:
Q . . . .
. . . . . . Q .
. . . . Q . . .

. . . . Q
. Q . . . . . .
. . . Q . . . .
. . . . . Q . .
. . Q . . . . .

Search Space Representation

Start
|
Place Queen in Column 1
|
Place Queen in Column 2
|
Place Queen in Column 3
|
Place Queen in Column 4
|
Place Queen in Column 5
|
Place Queen in Column 6
|
Place Queen in Column 7
|
Place Queen in Column 8
|
Goal State

If any placement is unsafe
|
Backtrack
|
Try another row

4. OLA Cab Booking Problem – Problem-Solving Agent

Problem Statement

A customer wants to travel from one location to another using the OLA Cab Booking mobile application. The customer can choose a cab type such as Mini, Micro, Sedan, Prime, Shared, etc., based on comfort, fare, and availability.

i. Type of Problem-Solving Agent

Goal-Based Problem-Solving Agent

Reason:

A Goal-Based Agent is the most suitable because:

- The customer's goal is to reach the destination.
- The agent searches for available cab options.
- It compares different cab types based on cost, comfort, availability, and estimated arrival time.
- Finally, it selects the cab that best satisfies the user's goal.

ii. Problem Formulation

Initial State

- Customer opens the OLA application.
- Enters pickup and destination locations.

Actions

- Enter pickup location.
- Enter destination.
- Search available cabs.
- Compare cab types.
- Select preferred cab.
- Confirm booking.
- Driver picks up customer.
- Customer reaches destination.

Goal State

- Customer successfully reaches the destination.

Path Cost

The cost can be evaluated based on:

- Fare
- Travel time
- Distance
- Waiting time

iii. Pseudocode

Algorithm: OLA Cab Booking

```
Start

Input Pickup_Location
Input Destination

Search Available_Cabs

Display:
  Mini
  Micro
  Sedan
  Prime
  Shared

Input User_Choice

IF Selected Cab is Available THEN
    Calculate Fare
    Display Estimated Arrival Time
    Confirm Booking

    IF Booking Confirmed THEN
        Assign Driver
        Driver Reaches Pickup Location
        Customer Boards Cab
        Travel to Destination
        Display "Trip Completed Successfully"
    ELSE
        Display "Booking Cancelled"
    ENDIF

ELSE
    Display "Selected Cab Not Available"
    Suggest Another Cab
ENDIF

Stop
```

Flow Diagram

```
        Start
          |
Open OLA App
          |
Enter Pickup Location
         |
Enter Destination
         |
Search Available Cabs
         |
Display Cab Types
(Mini, Micro, Sedan, Prime, Shared)
        |
Select Cab
        |
Is Cab Available?
         |
 ┌──────────Yes───────────────────────┐
     |                |
  Calculate Fare     No
    |                |
Confirm Booking   Show Other Cabs
         |
Assign Driver
         |
Travel to Destination
        |
Trip Completed
       |
Stop
```

Performance Measure

The agent's performance can be evaluated using:

- Correct cab allocation
- Minimum waiting time
- Shortest travel time
- Lowest fare

- Customer satisfaction
- Successful trip completion

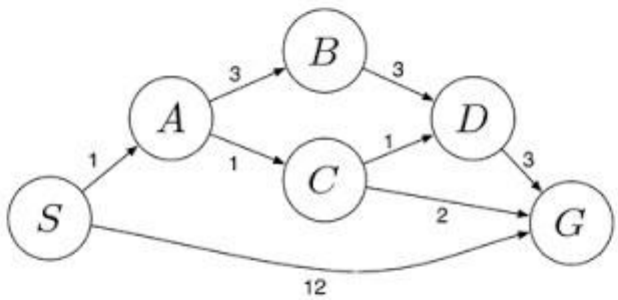
 Problem-Solving Agent: Goal-Based Agent

 Reason: The agent selects the best available cab to achieve the customer's goal of reaching the destination efficiently based on cost, comfort, availability, and travel time.

 Goal State: Customer reaches the destination successfully.

 Pseudocode: Use the algorithm provided above.

5. Uniform Cost Search (UCS) for Logistics Delivery Network
Algorithm: Uniform Cost Search (UCS)
Initial State: Starting warehouse **S**



Goal State: Destination warehouse **G**
Step 1: Start from source node **S** with path cost **0**.
Step 2: Insert **S** into the priority queue.
Step 3: Remove the node with the **lowest cumulative path cost**.
Step 4: If the removed node is the **goal node (G)**, stop and return the path.
Step 5: Otherwise, expand the node and add all its neighboring nodes with their cumulative costs into the priority queue.
Step 6: Repeat Steps 3–5 until the destination is reached.

Pseudocode
Algorithm UCS(Graph, Start, Goal)

1. Create a Priority Queue PQ
2. Insert (Start, Cost = 0) into PQ
3. Mark all nodes as unvisited

```
4. While PQ is not empty
   Remove node with minimum cost

   If node = Goal
      Return Path and Total Cost

   If node is not visited
      Mark node as visited

      For each neighboring node
         NewCost = CurrentCost + EdgeCost
         Insert (Neighbor, NewCost) into PQ

5. Return "No Path Found"
```

Working Principle

1. Begin at the source warehouse **S**.
2. Expand the node with the **lowest travel cost**.
3. Continue exploring the least-cost path first.
4. Repeat until warehouse **G** is reached.
5. The first time **G** is selected from the priority queue, the path found is the **minimum-cost path**.

Advantages of UCS

- Always finds the **optimal (least-cost) path**.
- Works well when route costs are different.
- Suitable for logistics, GPS navigation, and route planning.

Applications

- Delivery route optimization
- GPS navigation
- Robot path planning
- Airline route planning
- Network routing